# Midterm project presentation

Quantized Neural-Network Acceleration on RV64IMC

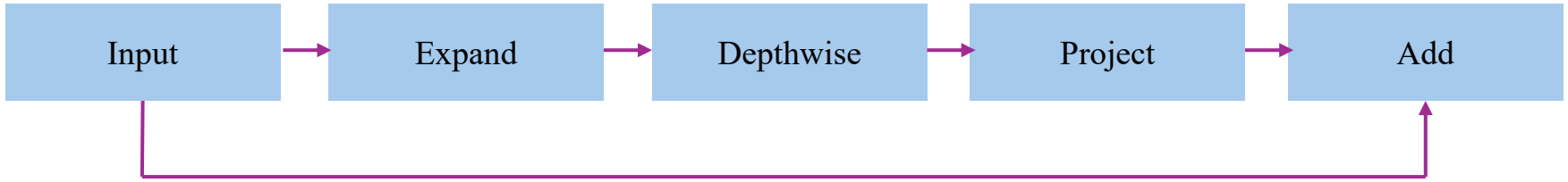
115061629 潘弘恩

# Basic guideline

- Evaluate the model and use custom ISA to speedup the model
- Use tiny change to get large performance improve

# Efficientnet-Lite0

- Image classification model from Google MediaPipe
- Fully integer with 64 ops
- A MBConv block:




Quantize uint→int8

Stem

1xMBConv1

2xMBConv6

2xMBConv6

3xMBConv6

3xMBConv6

4xMBConv6

1xMBConv6

Head 1x1 conv

Avg pool,Reshape
Fully connected
Softmax
Quantize

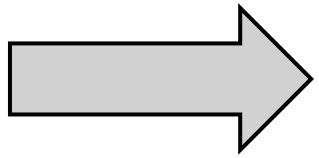
Source:https://huggingface.co/ketiswp/mediapipe-EfficientNet-Lite0-ImageNet-224-int8-uint8-onnx

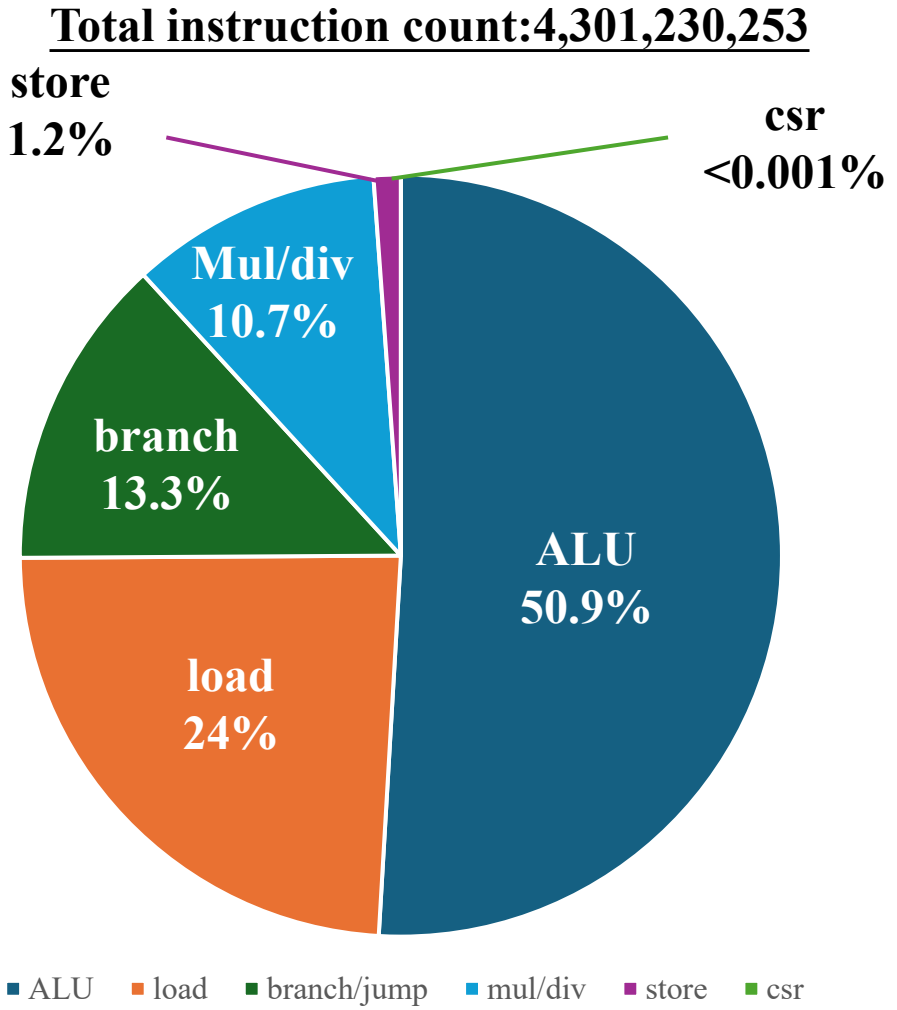
# Spike evaluation

Model code (c file) → riscv64-unknown-elf-gcc→spike -g

| op | type | instret | cycles | MAC/inst*100 | check |
|---|---|---|---|---|---|
| 0 | QUANTIZE | 7477124 | 7477124 | 0 | OK |
| 1 | CONV_2D | 209824520 | 209824520 | 5 | OK |
| 2 | DWCONV | 106118935 | 106118935 | 3 | OK |
| 3 | CONV_2D | 77190424 | 77190424 | 8 | OK |
| 4 | CONV_2D | 307227466 | 307227466 | 6 | OK |
| 5 | DWCONV | 79394204 | 79394204 | 3 | OK |
| 6 | CONV_2D | 67454638 | 67454638 | 10 | OK |
| 7 | CONV_2D | 143819902 | 143819902 | 7 | OK |
| 8 | DWCONV | 118153235 | 118153235 | 3 | OK |
| 9 | CONV_2D | 96358806 | 96358806 | 11 | OK |
| 10 | ADD | 8561557 | 8561557 | 0 | OK |
| 11 | CONV_2D | 143811310 | 143811310 | 7 | OK |
| 12 | DWCONV | 58776753 | 58776753 | 4 | OK |
| 13 | CONV_2D | 40139123 | 40139123 | 11 | OK |
| 14 | CONV_2D | 84019276 | 84019276 | 8 | OK |
| 15 | DWCONV | 95262886 | 95262886 | 4 | OK |
| 16 | CONV_2D | 64223103 | 64223103 | 11 | OK |
| 17 | ADD | 3575617 | 3575617 | 0 | OK |
| 18 | CONV_2D | 83854372 | 83854372 | 8 | OK |
| 19 | DWCONV | 12246994 | 12246994 | 3 | OK |
| 20 | CONV_2D | 32108479 | 32108479 | 11 | OK |
| 21 | CONV_2D | 72093640 | 72093640 | 10 | OK |
| 22 | DWCONV | 23745909 | 23745909 | 3 | OK |
| 23 | CONV_2D | 62214206 | 62214206 | 12 | OK |
| 24 | ADD | 1786861 | 1786861 | 0 | OK |
| 25 | CONV_2D | 72061269 | 72061269 | 10 | OK |
| 26 | DWCONV | 23728111 | 23728111 | 3 | OK |
| 27 | CONV_2D | 62214181 | 62214181 | 12 | OK |
| 28 | ADD | 1786873 | 1786873 | 0 | OK |
| 29 | CONV_2D | 72087038 | 72087038 | 10 | OK |
| 30 | DWCONV | 45360924 | 45360924 | 5 | OK |
| 31 | CONV_2D | 87098598 | 87098598 | 12 | OK |
| 32 | CONV_2D | 134474871 | 134474871 | 10 | OK |
| 33 | DWCONV | 63353265 | 63353265 | 5 | OK |
| 34 | CONV_2D | 120817045 | 120817045 | 12 | OK |
| 35 | ADD | 2499091 | 2499091 | 0 | OK |
| 36 | CONV_2D | 134426259 | 134426259 | 10 | OK |

The report using spike -g





# Custom ISA

X1

**qmac8 rd, rs1, rs2**

rs1:activation int8
rs2:weight int8
rd :accumulator int32 →int64

64 bit

rs1: | x7 | x6 | x5 | x4 | x3 | x2 | x1 | x0 |
rs2: | w7| w6| w5| w4 w3| w2| w1| w0 |

$$rd = rd + \sum xi * wi$$

X2

**qrequant rd, rs1, rs2**

rs1:accumulator result int32
rs2:requant parameters int8
rd :requant result int8

rs2: | act_max | act_min | out_zp | shift | mult (32 bits) |

$$rd = clamp(round(rs1, M) + out_{zp}, act_{min}, act_max)$$

$$where\ M = \frac{mult}{2^{31}} * 2^{Shift}$$

# Custom ISA

X1

**qmac8 rd, rs1, rs2**

rs1:activation int8
rs2:weight int8
rd :accumulator int32 →int64

64 bit

rs1: | x7 | x6 | x5 | x4 | x3 | x2 |
rs2: | w7| w6| w5| w4 w3| w2| w

$$rd = rd + \sum xi * wi$$

**Focusing on 1x1 conv**

X2

**ant rd, rs1, rs2**

or result int32
meters int8
int8

min | out_zp | shift | mult (32 bits) |

$d(rs1, M) + out_{zp}, act_{min}, act_max)$

$$\frac{alt}{2^{31}} * 2^{Shift}$$

# Why 1x1 conv

1.Dominates the model computation

-87.9% of MACs, 75.5% of B0 instructions

2.It’s a simple general matrix multiply

-fit xqmac8 describtion in project.md

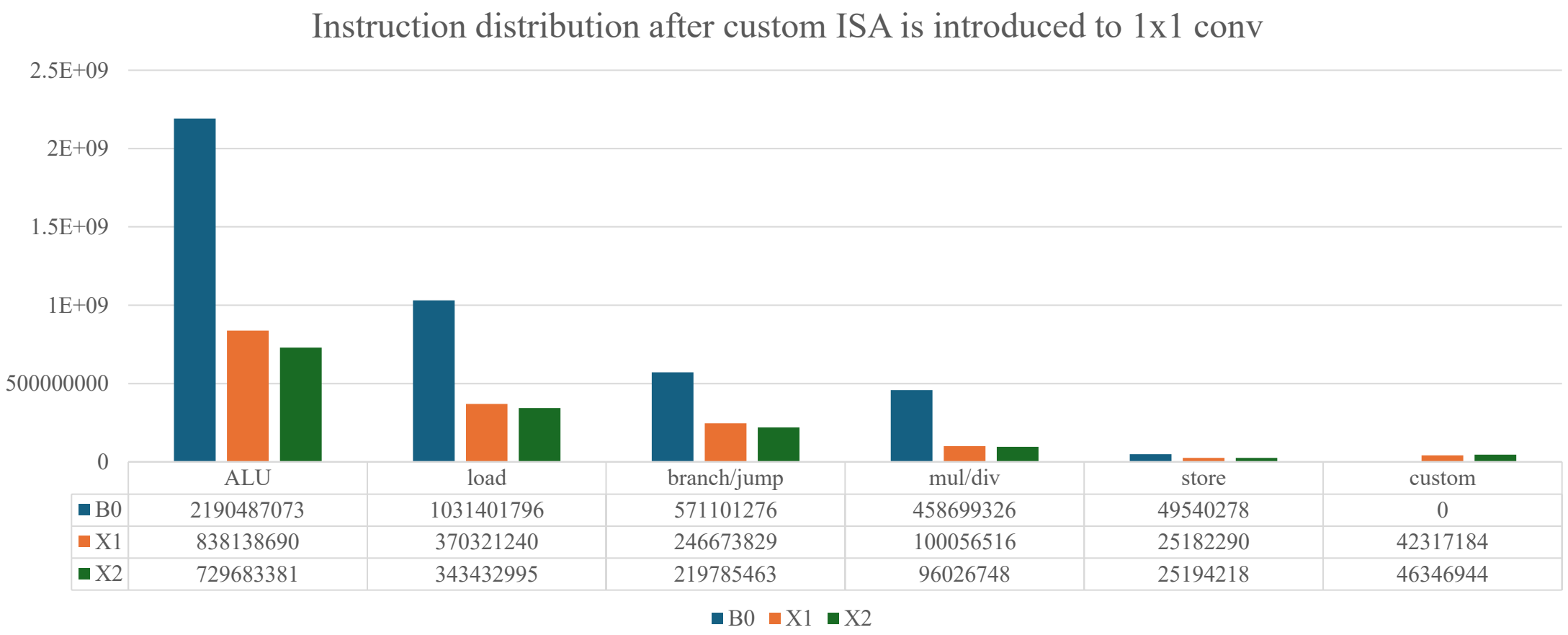
3.Data layout fits naturally

-Cin of all 32 layers is a multiples of 8

-8 activation/weight are contiguous→ no need data rearrangement

-stride = 1, no padding → straightforward

# After introducing custom ISA




| | ALU | load | branch/jump | mul/div | store | custom |
|---|---|---|---|---|---|---|
| B0 | 2190487073 | 1031401796 | 571101276 | 458699326 | 49540278 | 0 |
| X1 | 838138690 | 370321240 | 246673829 | 100056516 | 25182290 | 42317184 |
| X2 | 729683381 | 343432995 | 219785463 | 96026748 | 25194218 | 46346944 |

Total instruction count:B0=4,301,230,253 X1=1,622,690,253 X2=1,460,470,253

# Performance improvement(by Instruction count)

| | B0 | X1 | X2 |
|---|---|---|---|
| 1x1 convolution (32 layers) | 3218M | **539M** | **377M** |
| Speedup vs B0 | 1.00x | 5.97x | 8.53x |
| Whole model (w/o softmax) | 4264M | 1585M | 1423M |
| Whole model speedup vs B0 | 1.00x | 2.69x | 3.00x |
| Depthwise % | 18.6% | 50.0% | 55.7% |
| Top-1 (golden = 208) | 208 | 208 | 208 |

*This table is based on kernel itself, and the data from previous pages include spike from start to end

# What to do next

1.Use the lookup table technique to add Softmax and build full model

2. Evaluate with ImageNet dataset (50000 image/10 image/1 image)

3.Base on full model to build CV-Wally extension